📘 jQuery Built-in Methods

jQuery provides **built-in methods** that operate on the *jQuery object* returned by a selector.

When you write:

$("#box")

You receive a **jQuery-wrapped collection**, and built-in methods operate on that collection.

We will classify them professionally:

1. Content Methods
2. CSS & Class Methods
3. DOM Manipulation Methods
4. Traversing Methods
5. Event Methods
6. Effects & Animation Methods
7. AJAX Methods
8. Utility Methods

1️⃣ Content Methods

Used to read or modify element content.

🔹 .text()

Gets or sets plain text.

$("#para").text("Hello Students");

If used without parameter → returns text.

var value = $("#para").text();

🔹 .html()

Gets or sets HTML content.

```
$("#para").html("<b>Bold Text</b>");
```

Difference:

- text() → plain text
- html() → renders HTML

🔹 .val()

Used for form elements.

```
$("#username").val();
$("#username").val("Admin");
```

🔹 .attr()

Get or set attribute.

```
$("#img").attr("src");
$("#img").attr("src", "new.png");
```

🔹 .prop()

Used for properties (checkbox, selected, disabled).

```
$("#check").prop("checked", true);
```

2️⃣ CSS & Class Methods

🔹 .css()

```
$("#box").css("color", "red");
```

Multiple styles:

```
$("#box").css({
  "background": "black",
```

```
  "color": "white"
});
```

- **.addClass()**

```
$("#box").addClass("highlight");
```

- **.removeClass()**

```
$("#box").removeClass("highlight");
```

- **.toggleClass()**

```
$("#box").toggleClass("highlight");
```

3️⃣ DOM Manipulation Methods

These modify structure.

- **.append()**

Adds inside element (end).

```
$("#list").append("<li>New Item</li>");
```

- **.prepend()**

Adds inside element (beginning).

```
$("#list").prepend("<li>First Item</li>");
```

- **.before()**

Adds before element.

```
$("#box").before("<p>Before Box</p>");
```

- **.after()**

Adds after element.

```
$("#box").after("<p>After Box</p>");
```

- **.remove()**

Removes element completely.

```
$("#box").remove();
```

- **.empty()**

Removes child elements only.

```
$("#box").empty();
```

4️⃣ Traversing Methods

Used to navigate DOM relationships.

- **.parent()**

```
$("#child").parent();
```

- **.children()**

```
$("#menu").children();
```

- **.find()**

```
$("#menu").find("li");
```

🔹 .siblings()

```
$("#item").siblings();
```

🔹 .next() and .prev()

```
$("#item").next();
$("#item").prev();
```

5️⃣ Event Methods

🔹 .click()

```
$("#btn").click(function(){
  alert("Clicked");
});
```

🔹 .on() (Recommended)

```
$("#btn").on("click", function(){
  alert("Clicked");
});
```

Supports dynamic elements.

🔹 .off()

Removes event handler.

```
$("#btn").off("click");
```

🔹 **Common Events**

- click
- dblclick
- change
- submit
- keyup
- keydown
- mouseenter
- mouseleave

6️⃣ Effects & Animation Methods

🔹 **.hide() / .show()**

```
$("#box").hide();
$("#box").show();
```

🔹 **.toggle()**

```
$("#box").toggle();
```

🔹 **.fadeIn() / .fadeOut()**

```
$("#box").fadeIn(1000);
```

🔹 **.slideUp() / .slideDown()**

```
$("#box").slideDown(1000);
```

🔹 .animate()

Custom animation.

```
$("#box").animate({
  left: "250px",
  opacity: 0.5
}, 1000);
```

7️⃣ AJAX Methods

Important for real-time applications.

🔹 $.get()

```
$.get("data.json", function(data){
  console.log(data);
});
```

🔹 $.post()

```
$.post("api.php", {name:"John"}, function(response){
  console.log(response);
});
```

🔹 $.ajax() (Most Powerful)

```
$.ajax({
  url: "data.json",
  type: "GET",
  success: function(data){
    console.log(data);
```

```
  }
});
```

8️⃣ Utility Methods

🔹 $.each()

Loop through elements or arrays.

```
$.each([1,2,3], function(index, value){
  console.log(value);
});
```

🔹 $.trim()

Removes whitespace.

```
$.trim("  hello  ");
```

🔹 $.extend()

Merge objects.

```
$.extend(obj1, obj2);
```

📊 Summary Table

Category	Methods
Content	text(), html(), val()
CSS	css(), addClass()
DOM	append(), remove()

Category	Methods
Events	on(), click()
Effects	hide(), fadeIn()
AJAX	$.get(), $.ajax()
Utility	$.each(), $.extend()